

Задачи по линейной алгебре с РМО

1. (1995) Let n, k be positive integers and $A_i \in M_n(\mathbb{R})$, $i = 1, \dots, k$. Prove that

$$\det \left(\sum_{i=1}^k A_i^T A_i \right) \geq 0.$$

2. Let $A \in M_{3,2}(\mathbb{C})$ and $B \in M_{2,3}(\mathbb{C})$ be such that

$$AB = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 0 & 2 \\ 0 & 0 & 2 \end{pmatrix}.$$

Calculate $\det(BA)$

3. (1996) Let $A, B \in M_2(\mathbb{R})$ such that $\det(AB + BA) \leq 0$. Prove that

$$\det(A^2 + B^2) \geq 0.$$

4. Let $A, B, C, D \in M_n(\mathbb{C})$, where A and C are invertible. Prove that if $A^k B = C^k D$ for every positive integer k , then $B = D$.

5. (1997) Let $m \geq 2$ and $n \geq 1$ be integers and $A = (a_{ij})$ a square matrix of order n with integer entries. Prove that for any permutation $\sigma \in S_n$ there exists a function $\varepsilon : \{1, 2, \dots, n\} \rightarrow \{0, 1\}$ such that replacing the entries $a_{\sigma(1)1}, a_{\sigma(2)2}, \dots, a_{\sigma(n)n}$ of A respectively by

$$a_{\sigma(1)1} + \varepsilon(1), \quad a_{\sigma(2)2} + \varepsilon(2), \quad \dots, \quad a_{\sigma(n)n} + \varepsilon(n),$$

the determinant of the matrix A_ε thus obtained is not divisible by m .

6. Let A be a square matrix of odd order (at least 3) whose entries are odd integers. Prove that if A is invertible, then it is not possible for all the minors of the entries of a row to have equal absolute values.
7. (1998) Consider nonzero matrices $A_0, A_1, \dots, A_n \in M_2(\mathbb{R})$, $n \geq 2$, with the properties: $A_0 \neq aI_2$ for any $a \in \mathbb{R}$ and $A_0 A_k = A_k A_0$ for all $k = 1, \dots, n$. Prove that

a) $\det \left(\sum_{k=1}^n A_k^2 \right) \geq 0$;

b) if $\det \left(\sum_{k=1}^n A_k^2 \right) = 0$ and $A_2 \neq aA_1$ for any $a \in \mathbb{R}$, then $\sum_{k=1}^n A_k^2 = O_2$.

8. (1999) Let $A \in M_2(\mathbb{C})$ and $C(A) = \{B \in M_2(\mathbb{C}) : AB = BA\}$. Prove that

$$|\det(A + B)| \geq |\det B|,$$

for any $B \in C(A)$, if and only if $A^2 = O_2$.

9. (2000) Let

$$M = \{A \in M_2(\mathbb{C}) \mid \det(A - \lambda I) = 0 \Rightarrow |\lambda| < 1\}.$$

Prove that:

$$X, Y \in M \wedge XY = YX \Rightarrow X - Y \in M.$$

10. (2001) Consider a matrix $A \in M_n(\mathbb{C})$ with rank r , where $n \geq 2$ and $1 \leq r \leq n - 1$.

- a) Show that there exist $B \in M_{n,r}(\mathbb{C})$ and $C \in M_{r,n}(\mathbb{C})$, with $\text{rank}(B) = \text{rank}(C) = r$, such that $A = BC$.
- b) Show that the matrix A satisfies a polynomial equation of degree $r + 1$ with complex coefficients.

11. (2002) Let $A \in M_4(\mathbb{C})$ be a non-zero matrix.

- a) If $\text{rank}(A) = r < 4$, prove that there exist two invertible matrices $U, V \in M_4(\mathbb{C})$ such that:

$$UAV = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix},$$

where I_r is the $r \times r$ identity matrix.

- b) Show that if A and A^2 have the same rank k , then the matrix A^n has rank k for any $n \geq 3$.

12. (2003) Let $1 \leq a_1 < a_2 < a_3 < a_4$ and $x_1 < x_2 < x_3 < x_4$ be eight real numbers. Prove that

$$\begin{vmatrix} a_1^{x_1} & a_1^{x_2} & a_1^{x_3} & a_1^{x_4} \\ a_2^{x_1} & a_2^{x_2} & a_2^{x_3} & a_2^{x_4} \\ a_3^{x_1} & a_3^{x_2} & a_3^{x_3} & a_3^{x_4} \\ a_4^{x_1} & a_4^{x_2} & a_4^{x_3} & a_4^{x_4} \end{vmatrix} > 0.$$

13. Let A be a 3×3 real matrix. Prove the following statements.

- a) $f(A) \neq O_3$ for any polynomial $f \in \mathbb{R}[X]$ whose roots are not real.
- b) $\exists n \in \mathbb{N} \left((A + \text{adj}(A))^n = A^n + (\text{adj}(A))^n \right) \Leftrightarrow \det(A) = 0$.

14. (2004) Let $n \in \mathbb{N}$, $n \geq 2$.

- (a) Give an example of two matrices $A, B \in M_n(\mathbb{C})$ such that

$$\text{rank}(AB) - \text{rank}(BA) = \left\lfloor \frac{n}{2} \right\rfloor.$$

- (b) Prove that for all matrices $X, Y \in M_n(\mathbb{C})$ we have

$$\text{rank}(XY) - \text{rank}(YX) \leq \left\lfloor \frac{n}{2} \right\rfloor.$$

15. (2005) Let $n \geq 2$ be a fixed integer. We call an $n \times n$ matrix A with rational entries a *radical matrix* if there exist infinitely many positive integers k such that the equation $X^k = A$ has solutions in the set of $n \times n$ matrices with rational entries.

- a) Prove that if A is a radical matrix then $\det A \in \{-1, 0, 1\}$ and there exist infinitely many radical matrices with determinant 1.
- b) Prove that there exist infinitely many matrices that are not radical and have determinant 0, and also infinitely many matrices that are not radical and have determinant 1.

16. Let $X_1, X_2, \dots, X_m$ be a numbering of the $m = 2^n - 1$ non-empty subsets of the set $\{1, 2, \dots, n\}$, $n \geq 2$. Consider the matrix $(a_{ij})_{1 \leq i, j \leq m}$ where

$$a_{ij} = \begin{cases} 0, & \text{if } X_i \cap X_j = \emptyset, \\ 1, & \text{otherwise.} \end{cases}$$

Prove that the determinant d of this matrix does not depend on the numbering and compute d .

17. (2006) Let A be an $n \times n$ matrix with complex elements and let $\text{adj}(A)$ be the classical adjoint (adjugate) of A . Prove that if there exists a positive integer m such that $(\text{adj}(A))^m = 0$, then $(\text{adj}(A))^2 = 0$.

18. We define a *pseudo-inverse* $B \in M_n(\mathbb{C})$ of a matrix $A \in M_n(\mathbb{C})$ as a matrix that satisfies the relations

$$A = ABA \quad \text{and} \quad B = BAB.$$

- a) Prove that any square matrix has at least one pseudo-inverse.
- b) For which matrices A is the pseudo-inverse unique?

19. (2007) Let $A, B \in M_2(\mathbb{R})$ be real 2×2 matrices that satisfy $A^2 + B^2 = AB$. Prove that

$$(AB - BA)^2 = O_2.$$

20. Let $n \geq 2$ be an integer and denote by H_n the set of column vectors $(x_1, x_2, \dots, x_n)^T \in \mathbb{R}^n$ such that

$$\sum_{i=1}^n |x_i| = 1.$$

Prove that there exist only finitely many matrices $A \in M_n(\mathbb{R})$ such that the linear map $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ given by $f(x) = Ax$ has the property $f(H_n) = H_n$.

Comment. In the contest, the problem was given with:

- (a) Prove the above for $n = 2$;
- (b) Prove the above for $n \geq 3$.

21. (2008) Let A be an $n \times n$ matrix with complex elements. Prove that $A^{-1} = A$ if and only if there exists an invertible matrix B with complex elements such that $A = B^{-1} \cdot B$.

22. Let $A = (a_{ij})_{1 \leq i, j \leq n}$ be a real $n \times n$ matrix such that $a_{ij} + a_{ji} = 0$ for all i, j . Prove that for all non-negative real numbers x, y we have

$$\det(A + xI_n) \cdot \det(A + yI_n) \geq (\det(A + \sqrt{xy}I_n))^2.$$

23. (2009) Let $A, B \in M_n(\mathbb{C})$ such that $AB = BA$ and $\det B \neq 0$.

- a) If $|\det(A + zB)| = 1$ for every $z \in \mathbb{C}$ with $|z| = 1$, then $A^n = O_n$.
- b) Does the statement in part (a) remain true if $AB \neq BA$?

24. (2010) Let $a, b \in \mathbb{R}$ such that $b > a^2$. Find all matrices $A \in \mathcal{M}_2(\mathbb{R})$ such that

$$\det(A^2 - 2aA + bI_2) = 0.$$

25. Let $A, B, C \in \mathcal{M}_n(\mathbb{R})$ be such that $ABC = O_n$ and $\text{rank } B = 1$. Prove that either $AB = O_n$ or $BC = O_n$.

26. (2011) A row of a matrix belonging to $M_n(\mathbb{C})$ is said to be *permutable* if no matter how we permute the entries of that row, the value of the determinant does not change. Prove that if a matrix has two permutable rows, then its determinant is equal to 0.

27. Let $A, B \in M_2(\mathbb{C})$ be such that $A^2 + B^2 = 2AB$.

- a) Prove that $AB = BA$.
- b) Prove that $\text{tr}(A) = \text{tr}(B)$.

28. (2012) Let n and k be two natural numbers such that $n \geq 2$ and $1 \leq k \leq n-1$. Prove that if the matrix $A \in \mathcal{M}_n(\mathbb{C})$ has exactly k minors of order $n-1$ equal to 0, then $\det(A) \neq 0$.

29. Let $A, B \in \mathcal{M}_4(\mathbb{R})$ be such that $AB = BA$ and $\det(A^2 + AB + B^2) = 0$. Prove that:

$$\det(A + B) + 3\det(A - B) = 6\det(A) + 6\det(B).$$

30. (2013) Given a non-invertible matrix A of order n with real entries, $n \geq 2$, and let A^* denote the adjugate matrix of A . Prove that

$$\text{tr}(A^*) \neq -1$$

if and only if the matrix $I_n + A^*$ is invertible.

31. Let m and n be natural numbers, $m, n \geq 2$. Consider matrices $A_1, A_2, \dots, A_m \in M_n(\mathbb{R})$ that are not all nilpotent. Prove that there exists an integer $k > 0$ such that

$$A_1^k + A_2^k + \dots + A_m^k \neq O_n.$$

32. (2014) Let $A, B \in M_n(\mathbb{C})$ be two square matrices satisfying $A^2 + B^2 = 2AB$.

- (1) Prove that $\det(AB - BA) = 0$.

- (2) If $\text{rank}(A - B) = 1$, then prove that $AB = BA$.
33. Let $A \in M_4(\mathbb{R})$ be an invertible matrix whose trace is equal to the trace of its adjugate, and this trace is nonzero. Show that $A^2 + I$ is singular if and only if there exists a nonzero matrix in $M_4(\mathbb{R})$ that anti-commutes with A .
34. (2015) Let A be a 5×5 complex matrix with trace 0 such that $I_5 - A$ is invertible. Prove that $A^5 \neq I_5$.
35. Let k, m, n be natural numbers, let A be an $m \times n$ matrix, B an $n \times m$ matrix, and let $a_0, a_1, \dots, a_k$ be complex numbers such that:
- (i) $m \geq n \geq 2$;
 - (ii) $a_0 I_m + a_1 AB + a_2 (AB)^2 + \dots + a_k (AB)^k = O_m$;
 - (iii) $a_0 I_n + a_1 BA + a_2 (BA)^2 + \dots + a_k (BA)^k \neq O_n$.

Prove that $a_0 = 0$.

36. (2016) Let A be a 2×2 real matrix such that

$$\det(A^d - I_2) = \det(A^d + I_2) \quad \text{for all } d \in \{2014, 2016\}.$$

Prove that

$$\det(A^n - I_2) = \det(A^n + I_2) \quad \text{for every natural number } n.$$

37. Consider a natural number $n \geq 2$ and three $n \times n$ complex matrices A, B, C such that A is invertible, B is obtained from A by replacing its first row with zeros, and C is obtained by taking the last $n - 1$ rows of A and placing them above a row of zeros (so that the first $n - 1$ rows of C are rows $2, 3, \dots, n$ of A , and the last row of C is zero). Prove that:
- a) $\text{rank}(A^{-1}B) = \text{rank}((A^{-1}B)^2) = \dots = \text{rank}((A^{-1}B)^n)$;
 - b) $\text{rank}(A^{-1}C) > \text{rank}((A^{-1}C)^2) > \dots > \text{rank}((A^{-1}C)^n)$.
38. (2017) Let $n \geq 2$ and k be natural numbers, and let $A_1, A_2, \dots, A_k$ be $n \times n$ symmetric real matrices. Then the following statements are equivalent:

- 1) $\left| \sum_{i=1}^k A_i^2 = 0 \right|$.
- 2) $\left| \sum_{i=1}^k A_i B_i = 0 \right|$, for all $B_1, B_2, \dots, B_k \in M_n(\mathbb{R})$.

39. Let $n \geq 2$ be a natural number and let A, B be two $n \times n$ complex matrices satisfying $(AB)^3 = O_n$. Does this imply that $(BA)^3 = O_n$?

40. (2018) Let $n \geq 2$ be a positive integer and, for all vectors with integer entries

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix},$$

let $\delta(X) \geq 0$ denote the greatest common divisor of $x_1, x_2, \dots, x_n$. Also, consider $A \in M_n(\mathbb{Z})$. Prove that the following statements are equivalent:

i) $|\det A| = 1$.

ii) $\delta(AX) = \delta(X)$ for all vectors $X \in M_{n,1}(\mathbb{Z})$.

41. Let n be an integer with $n > 2$ and let $A \in M_n(\mathbb{C})$ such that $\text{rank } A \neq \text{rank } A^2$. Prove that there exists a nonzero matrix $B \in M_n(\mathbb{C})$ such that

$$AB = BA = B^2 = 0.$$

42. (2019) Let $n > 2$ and $A, B \in M_n(\mathbb{C})$ be such that there exists an idempotent matrix $C \in M_n(\mathbb{C})$ (i.e., $C^2 = C$) for which $\text{adj}(C) = AB - BA$. Prove that

$$(AB - BA)^2 = 0.$$

(Here $\text{adj}(X)$ denotes the adjugate matrix of X .)

43. Let p be a prime number. For any permutation $\sigma \in S_p$ (the symmetric group on $\{1, 2, \dots, p\}$), define the matrix $A_\sigma = (a_{ij}) \in M_p(\mathbb{Z})$ by

$$a_{ij} = \sigma^{i-1}(j), \quad i, j = 1, \dots, p,$$

where σ^0 is the identity permutation and $\sigma^k = \sigma \circ \sigma \circ \dots \circ \sigma$ (k times). Prove that the set

$$D = \{ |\det A_\sigma| : \sigma \in S_p \}$$

has at most $1 + (p-2)!$ elements.

44. (2021) Let $n > 2$ and $a_1, a_2, \dots, a_n$ be nonzero real numbers, not necessarily distinct. Define the matrix $A = (a_{ij})_{1 \leq i, j \leq n} \in M_n(\mathbb{R})$ by $a_{ij} = \max\{a_i, a_j\}$ for all $i, j \in \{1, 2, \dots, n\}$. Show that

$$\text{rank}(A) = \text{card}\{a_k \mid k = 1, 2, \dots, n\},$$

where “card” denotes the number of distinct elements.

45. Let $n \geq 2$ and let $A, B \in M_n(\mathbb{R})$. Suppose there exists $x \in \mathbb{R} \setminus \{0, 1\}$ such that

$$xAB + (1-x)BA = I_n.$$

Prove that

$$(AB - BA)^n = O_n.$$

46. (2022) Let $\mathcal{F}$ be the set of pairs of matrices $(A, B) \in \mathcal{M}_2(\mathbb{Z}) \times \mathcal{M}_2(\mathbb{Z})$ for which there exists some positive integer k and matrices $C_1, C_2, \dots, C_k \in \{A, B\}$ such that $C_1 C_2 \cdots C_k = O_2$. For each $(A, B) \in \mathcal{F}$, let $k(A, B)$ denote the minimal positive integer k with this property.

a) Let $(A, B) \in \mathcal{F}$ with $\det(A) = 0$, $\det(B) \neq 0$ and $k(A, B) = p + 2$ for some $p \in \mathbb{N}^*$. Show that

$$AB^p A = O_2.$$

b) Prove that for any $k \geq 3$ there exists a pair $(A, B) \in \mathcal{F}$ such that $k(A, B) = k$.

47. Let $A, B \in M_n(\mathbb{C})$ such that $A^2 + B^2 = 2AB$. Prove that for any complex number α ,

$$\det(A - \alpha I_n) = \det(B - \alpha I_n).$$

48. (2023) Let $A, B \in M_n(\mathbb{R})$. Show that $\text{rank}(A) = \text{rank}(B)$ if and only if there exist nonsingular matrices $X, Y, Z \in M_n(\mathbb{R})$ such that

$$AX + YB = AZB.$$

49. Let n be a natural number, $n \geq 2$, and let $A, B \in M_n(\mathbb{C})$ be matrices satisfying $A^2 B = A$.

a) Prove that $(AB - BA)^2 = O_n$.

b) Show that for every natural number k with $k \leq \frac{n}{2}$, there exist matrices $A, B \in M_n(\mathbb{C})$ satisfying the given property such that $\text{rank}(AB - BA) = k$.

50. (2024) Let $A \in \mathcal{M}_n(\mathbb{R})$ be an invertible matrix.

a) Prove that the eigenvalues of AA^T are positive real numbers.

b) Assume that there exist two distinct positive integers p and q such that $(AA^T)^p = (A^T A)^q$. Prove that $A^T = A^{-1}$.

51. Let $A, B \in \mathcal{M}_n(\mathbb{R})$. Consider the function $f : \mathcal{M}_n(\mathbb{C}) \rightarrow \mathcal{M}_n(\mathbb{C})$ defined by

$$f(Z) = AZ + B\overline{Z}, \quad Z \in \mathcal{M}_n(\mathbb{C}),$$

where $\overline{Z}$ is the matrix whose entries are the complex conjugates of the corresponding entries of Z . Prove that the following statements are equivalent:

- (1) the function f is injective;
- (2) the function f is surjective;
- (3) the matrices $A + B$ and $A - B$ are invertible.

52. (2025) Let n be a positive integer, and a, b be two complex numbers such that $a \neq 1$ and $b^k \neq 1$ for any $k \in \{1, 2, \dots, n\}$. Suppose matrices $A, B \in M_n(\mathbb{C})$ satisfy the relation

$$BA = aI_n + bAB.$$

Prove that A and B are invertible.

53. Let $A, B \in M_n(\mathbb{C})$ be two matrices such that

$$A + B = AB + BA.$$

Prove that:

- a) if n is odd, then $\det(AB - BA) = 0$;
- b) if $\operatorname{tr}(A) \neq \operatorname{tr}(B)$, then $\det(AB - BA) = 0$.

54. (2026) Let $m, n \in \mathbb{N}^+$. Consider a matrices $A, B \in \mathcal{M}_n(\mathbb{C})$ and complex numbers $a_0, a_1, \dots, a_m$ with $a_0 \neq 0$ and $a_m \neq 0$ such that

$$A \cdot (a_0 E + a_1 B + a_2 B^2 + \dots + a_m B^m) = B.$$

Prove that $AB = BA$.

55. Let $M = \{A_1, A_2, \dots, A_n\}$ be a set of $n \geq 2$ matrices in $\mathcal{M}_{2026}(\mathbb{C})$ such that $AB \in M$ for all $A, B \in M$. Prove that there exist $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n \in \{-1, 0, 1\}$, not all zero, such that

$$\operatorname{rank}(\varepsilon_1 A_1 + \varepsilon_2 A_2 + \dots + \varepsilon_n A_n) \leq 1013.$$